

Performance-Enhanced Robust Iterative Learning Control With Experimental Application to PMSM Position Tracking

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Abstract—This brief develops an innovative robust iterative learning control law using the repetitive process setting. The new design is experimentally validated through a comprehensive set of experiments highlighting the capabilities for the position tracking control of a permanent magnet synchronous motor subject to load disturbances in the presence of uncertainties in selected parameters.

Index Terms—Iterative learning control (ILC), permanent magnet synchronous motor (PMSM), repetitive process stability theory.

I. INTRODUCTION

ITERATIVE learning control (ILC) was especially developed for systems or processes that repeat the same finite duration operation over and over again, such as gantry robots in mass production facilities. Each repetition is termed a trial and its duration is the trial length. Once a trial is completed, all information generated are available for updating the control input so that the output follows the specified reference trajectory, including the cases where there is uncertainty in the plant model and/or load disturbances are present. The notation used in this brief is of the form $h_k(p)$, $0 \leq p \leq \alpha - 1$, where h is a scalar or vector-valued function under consideration, the nonnegative integer k denotes the trial number, and $\alpha < \infty$ is the number of samples along the trial (α times the constant sampling period gives the trial length).

Let $y^{\text{ref}}(p)$ denote the supplied reference trajectory. Then, the tracking error on trial k is $e_k(p) = y^{\text{ref}}(p) - y_k(p)$ where $y_k(p)$ is the output on trial k and the objective is the design of a controller to ensure that $e_k(p)$ converges to zero $\forall p$, $0 \leq p \leq \alpha - 1$ in k and the control input signals converge to

a so-called learned control. The first work on the ILC is widely credited to [1], and since then, it has been an expanding area of research in control systems with an ever widening range of applications. Starting points for the literature are the survey papers [2], [3].

In the case of discrete linear dynamics, there are many settings for the ILC design. In this brief, the repetitive process setting is used, which has already seen many designs reported [4] with experimental validation. However, repetitive process-based designs come at a price in terms of data storage required that may be problematic in some applications. This brief develops a new design where the storage of the previous trial state vector over the complete trial length is not required. It also introduces compensating countermeasures for other undesirable effects.

The new contributions in this brief are: 1) an integral action to compensate for unknown constant or slowly varying disturbances on the trial, where the compensating control action is applied on the trial in which they first appear, rather than on subsequent trials in other designs; 2) a feed-forward control action to reduce initial tracking errors in the early learning phase; and 3) allowing for plants with relative degree greater than unity unlike the alternative design in [5].

Model uncertainty described by the convex polytope is included in the design. Moreover, the merits of the new design are illustrated by applying the control strategy to a permanent magnet synchronous motor (PMSM). Finally, a comparison with competing repetitive process-based ILC designs is given.

In this brief, bold capital letters denote constant or parameter-dependent matrices or vectors. The zero and identity matrices with compatible dimensions are denoted by $\mathbf{0}$ and $\mathbf{I}$, respectively. A real symmetric positive (semi-) definite [resp. negative (semi-) definite] matrix is written as $\mathbf{Z} > 0$ (≥ 0) [< 0 (≤ 0)]. The symbol $(*)$ denotes block entries in symmetric matrices and $\mathbb{1}(p)$ denotes the unit step function.

II. OVERALL CONTROL CONCEPT

The new ILC design in this brief is based on a discrete-time linear parameter-dependent state-space plant model written in the ILC setting as

$$\begin{aligned} x_{k+1}(p+1) &= \mathbf{A}(\lambda)x_{k+1}(p) + \mathbf{B}(\lambda)(u_{k+1}(p) - d_{k+1}(p)) \\ y_{k+1}(p) &= \mathbf{C}x_{k+1}(p) \end{aligned} \quad (1)$$

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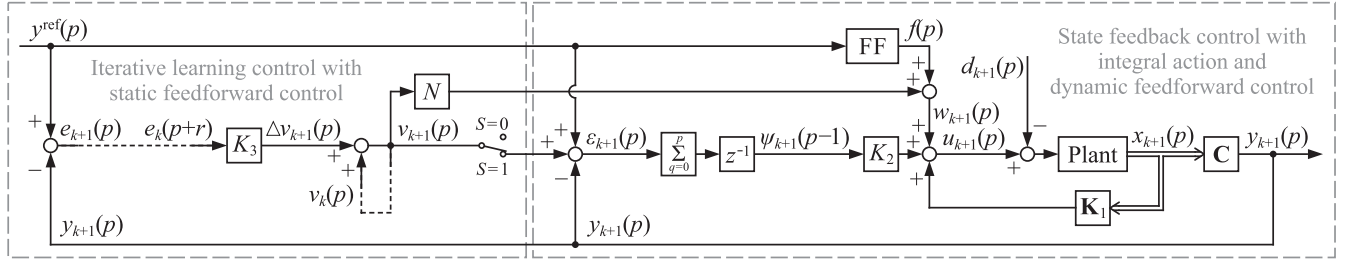


Fig. 1. ILC structure based on a state feedback controller with integral action and a dynamic feedforward controller.

where the matrices $\mathbf{A}(\lambda)$ and $\mathbf{B}(\lambda)$ are assumed to belong to a convex polytope described by

$$\begin{aligned} [\mathbf{A}(\lambda) \quad \mathbf{B}(\lambda)] &= \sum_{j=1}^M \lambda_j [\mathbf{A}_j \quad \mathbf{B}_j] \\ \lambda &= \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_M \end{bmatrix}, \quad \lambda_j \geq 0, \quad \sum_{j=1}^M \lambda_j = 1. \end{aligned} \quad (2)$$

Considering trial $k+1$, $x_{k+1}(p) \in \mathbb{R}^n$ denotes the state vector, $u_{k+1}(p) \in \mathbb{R}$ denotes the control input, $d_{k+1}(p) \in \mathbb{R}$ denotes the load disturbance input, and $y_{k+1}(p) \in \mathbb{R}$ denotes the output. The output vector $\mathbf{C}$ is assumed to be fixed and known exactly, and the plant relative degree is $r \geq 1$. This brief considers single-input single-output (SISO) systems, but the results can be generalized to the multi-input multi-output (MIMO) case.

In the repetitive process setting, the ILC law is of the form

$$u_{k+1}(p) = u_k(p) + \mathbf{K}_{\text{stab}}(x_{k+1}(p) - x_k(p)) + K_{\text{ILC}}e_k(p+1) \quad (3)$$

where the second term provides stabilizing control action and the third term is the ILC feedforward using the sample $p+1$ of the error information from the previous trial to improve the tracking accuracy during the current trial. The design of this control law is based on the linear repetitive process stability theory, and the associated controller gain matrices are computed using linear matrix inequalities (LMIs). The new ILC scheme developed in this brief is shown in Fig. 1 and adds the following features relative to previous designs.

- 1) In standard ILC designs, the tracking error for a system subject to a load disturbance is reduced only after starting the next trial [5]. In the new design, the impact of such a disturbance is compensated on the trial where it appears first and, hence, a possibly significant tracking error is avoided.
- 2) Large tracking errors occurring in the early trials or directly after a change in the reference trajectory are reduced.
- 3) An ILC design for uncertain plant models with relative degree $r > 1$ becomes possible.
- 4) The ILC algorithm implementation does not require additional memory to store $\mathbf{K}_{\text{stab}}x_k(p)$ in (3).

The term d_{k+1} in Fig. 1 represents a load disturbance entering on trial $k+1$, resulting in a steady-state tracking error in the dynamics along the trial (the response along a trial is

described in terms of the discrete-time variable p). This error can be reduced by the ILC law (3), but this compensation will only take effect on the next trial after the disturbances have occurred. One suitable countermeasure was given in [5], where the tracking error caused by disturbances is reduced by a disturbance observer. The stability of such an observer-based ILC can be proven after the design is complete [5]. This verification becomes redundant for the new design in this brief.

Another way of immediately reducing the tracking error caused by unknown constant or slowly varying disturbances, within the trial where they first appear, is to include integral feedback of the current tracking error. The implementation of this action is achieved by the output of the K_2 block in Fig. 1. Moreover, a disturbance observer also cannot remove this error completely due to its limited dynamics. It is reduced, however, to zero by the ILC action on subsequent trials. To increase the convergence speed, the ILC signal $v_{k+1}(p)$ acts on the closed-loop (state feedback) system by both the static feedforward gain N and the integral action.

In an ILC implementation, large tracking errors can arise during early trials or directly after a change in the reference trajectory [5]. One cause for errors in the initial trials is the assumption that the initial control signal is zero [$u_0(p) = 0$]. Errors after a change in the reference trajectory may arise, among other reasons, from the fact that stored information on the previous control input [$u_k(p)$] is outdated after this change and may even prevent accurate tracking. The control design developed in this brief aims to compensate for this effect by including a dynamic feedforward control. It takes advantage of the reference trajectory $y^{\text{ref}}(p)$ and is implemented within the FF block in Fig. 1, and the ILC update is given by $K_3e_k(p+r)$, building on a previous analysis in [4].

III. ROBUST STATE FEEDBACK WITH INTEGRAL ACTION AND DYNAMIC FEEDFORWARD CONTROL DESIGN

The proposed ILC scheme of Fig. 1 (switch S is in position 1) is based on a state feedback controller with integral action as well as both static and dynamic feedforward controllers. The design of these controllers is detailed in this section.

The integral action in the state feedback controller is introduced through

$$\psi_{k+1}(p) = \sum_{q=0}^p e_{k+1}(q) \quad (4)$$

with $\varepsilon_{k+1}(p) = y^{\text{ref}}(p) + v_{k+1}(p) - y_{k+1}(p)$. For design purposes, the signals $v_{k+1}(p)$, $y^{\text{ref}}(p)$, and $d_{k+1}(p)$ are set to zero as they do not influence stability. Applying the control law

$$u_{k+1}(p) = \mathbf{K}_1 x_{k+1}(p) + K_2 \psi_{k+1}(p-1) \quad (5)$$

to the plant model (1) leads to an augmented system representation that can be written as

$$\begin{aligned} \chi_{k+1}(p+1) &= (\mathbf{A}_s(\lambda) + \mathbf{B}_s(\lambda)\mathbf{K}_s)\chi_{k+1}(p) \\ y_{k+1}(p) &= \mathbf{C}_{\text{cl}}\chi_{k+1}(p) \end{aligned} \quad (6)$$

where

$$\begin{aligned} \mathbf{A}_s(\lambda) &= \begin{bmatrix} \mathbf{A}(\lambda) & \mathbf{0} \\ -\mathbf{C} & 1 \end{bmatrix}, \quad \chi_{k+1}(p) = \begin{bmatrix} x_{k+1}(p) \\ \psi_{k+1}(p-1) \end{bmatrix} \\ \mathbf{B}_s(\lambda) &= \begin{bmatrix} \mathbf{B}(\lambda) \\ 0 \end{bmatrix}, \quad \mathbf{C}_s = [\mathbf{C} \quad 0], \quad \mathbf{K}_s = [\mathbf{K}_1 \quad K_2]. \end{aligned} \quad (7)$$

The design of the state feedback law with the integral action requires $\mathbf{K}_s$ to be chosen such that (6) is stable for all possible $[\mathbf{A}(\lambda) \quad \mathbf{B}(\lambda)]$. One way of completing this design is to find $u_{k+1}(p) = \mathbf{K}_s \chi_{k+1}(p)$ that minimizes the cost function

$$J_{\text{cl}} = \sum_{p=0}^{\infty} (\chi_{k+1}^T(p) \mathbf{Q} \chi_{k+1}(p) + R u_{k+1}^2(p)) \quad (8)$$

where $\mathbf{Q} > 0$ and $R > 0$ are, respectively, an appropriately chosen weighting matrix and scalar factor. Many ways of solving this problem are known, and in this brief, the guaranteed cost method [6] is used.

For given $\chi_{k+1}(0) = \boldsymbol{\gamma}$, suppose that there exist compatibly dimensioned matrices $\mathbf{Y} > 0$, $\mathbf{W}$, and a positive scalar β such that for $j = 1, \dots, M$, the LMIs

$$\begin{bmatrix} \mathbf{Y} & (*) & (*) & (*) \\ \mathbf{A}_{sj}\mathbf{Y} + \mathbf{B}_{sj}\mathbf{W} & \mathbf{Y} & (*) & (*) \\ \mathbf{Y} & \mathbf{0} & \mathbf{Q}^{-1} & (*) \\ \mathbf{W} & \mathbf{0} & \mathbf{0} & \frac{1}{R} \end{bmatrix} \geq 0 \quad (9)$$

$$\begin{bmatrix} \beta & \boldsymbol{\gamma}^T \\ \boldsymbol{\gamma} & \mathbf{Y} \end{bmatrix} \geq 0 \quad (10)$$

are feasible, where

$$\mathbf{A}_{sj} = \begin{bmatrix} \mathbf{A}_j & \mathbf{0} \\ -\mathbf{C} & 1 \end{bmatrix}, \quad \mathbf{B}_{sj} = \begin{bmatrix} \mathbf{B}_j \\ 0 \end{bmatrix}. \quad (11)$$

Then (for the proof, see [6]), the following optimization procedure can be applied:

$$\min(\beta), \quad \text{subject to: } \mathbf{Y} > 0, \quad (9), \quad (10) \quad (12)$$

for $j = 1, \dots, M$. If a solution exists to this problem, the stabilizing gain vector is given by

$$\mathbf{K}_s = \mathbf{W}\mathbf{Y}^{-1} \quad (13)$$

where $\mathbf{K}_1$ and K_2 are obtained from the partitioning of $\mathbf{K}_s$ according to (7).

The next step is to design the dynamic and static feedforward control terms in Fig. 1. Using the dynamic feedforward controller (FF), accurate tracking can be obtained only if both the plant model and disturbances are exactly known. In the case of an uncertain system model and unknown—but

repetitive—disturbances, the dynamic feedforward controller only calculates an approximation to the control signal, which is enhanced by the ILC signal $v_{k+1}(p)$ computed using the K_3 block in Fig. 1. This design can be completed as detailed in Section V.

Also the static feedforward controller represented by block N in Fig. 1 is added to avoid instability arising due to excessive integration, which may lead to an integrator windup in the worst case. Under the assumption that $K_2 = 0$, $f(p) = 0$, and $d_{k+1}(p) = 0$, the resulting closed-loop transfer function with the state feedback control law applied has the scalar input, the signal $w_{k+1}(p) = N v_{k+1}(p)$ in Fig. 1. To make the dc gain of the series connection of block N and the closed-loop transfer function equal to one, the static feedforward gain N has to be chosen as the inverse of the dc gain of the feedback loop.

IV. REPETITIVE PROCESS FORMULATION AND ROBUST ILC DESIGN

The ILC signal $v_{k+1}(p)$ acts on the closed-loop state feedback control system as shown in Fig. 1 ($S = 1$). Setting $y^{\text{ref}}(p) = 0$ and $d_{k+1}(p) = 0$, the contribution of the signal $v_{k+1}(p)$ to the plant output is described by the state-space model

$$\begin{aligned} \chi_{k+1}(p+1) &= \mathbf{A}_{\text{cl}}(\lambda)\chi_{k+1}(p) + \mathbf{B}_{\text{ILC}}(\lambda)v_{k+1}(p) \\ y_{k+1}(p) &= \mathbf{C}_{\text{cl}}\chi_{k+1}(p) \end{aligned} \quad (14)$$

where

$$\begin{aligned} \mathbf{A}_{\text{cl}}(\lambda) &= \mathbf{A}_s(\lambda) + \mathbf{B}_s(\lambda)\mathbf{K}_s \\ \mathbf{B}_{\text{ILC}}(\lambda) &= \begin{bmatrix} \mathbf{B}(\lambda)N \\ 1 \end{bmatrix}. \end{aligned} \quad (15)$$

The ILC law provides the signal to be applied on the next trial, which is the sum of the signal used on the previous trial and a correction term, i.e.,

$$v_{k+1}(p) = v_k(p) + \Delta v_{k+1}(p) \quad (16)$$

with

$$\Delta v_{k+1}(p) = K_3 e_k(p+r) \quad (17)$$

in contrast to other repetitive process-based designs [5].

Introduce, for analysis purposes only, the vector

$$\eta_{k+1}(p) = \chi_{k+1}(p) - \chi_k(p). \quad (18)$$

Then, the application of (16) and (17) to (14) gives the following state-space model of the controlled ILC dynamics:

$$\begin{aligned} \eta_{k+1}(p+1) &= \mathbf{A}_{\text{cl}}(\lambda)\eta_{k+1}(p) + \mathbf{B}_{\text{ILC}}(\lambda)K_3 e_k(p+r) \\ e_{k+1}(p) &= -\mathbf{C}_{\text{cl}}\eta_{k+1}(p) + e_k(p). \end{aligned} \quad (19)$$

By applying the z -transform to (19), one obtains (see, e.g., the relevant references in [3] for the justification of why applying the z -transform to a finite trial length does not affect the final result)

$$\frac{e_{k+1}(z)}{e_k(z)} = -\mathbf{C}_{\text{cl}} z^r (z\mathbf{I} - \mathbf{A}_{\text{cl}}(\lambda))^{-1} \mathbf{B}_{\text{ILC}}(\lambda) K_3 + 1. \quad (20)$$

Moreover, see [4], this leads to the state-space model

$$\begin{aligned}\eta_{k+1}(p+1) &= \mathbf{A}_{cl}(\lambda)\eta_{k+1}(p) + \mathbf{B}_{ILC}(\lambda)K_3e_k(p) \\ e_{k+1}(p) &= -\Gamma(\lambda)\eta_{k+1}(p) + (1 - \Upsilon(\lambda)K_3)e_k(p)\end{aligned}\quad (21)$$

with

$$\begin{aligned}\Gamma(\lambda) &= \mathbf{C}_{cl}(\mathbf{A}_{cl}(\lambda))^r \\ \Upsilon(\lambda) &= \mathbf{C}_{cl}(\mathbf{A}_{cl}(\lambda))^{r-1}\mathbf{B}_{ILC}(\lambda).\end{aligned}\quad (22)$$

The set with vertices $[\mathbf{A}_{clj} \ \mathbf{B}_{ILCj} \ \Gamma_j \ \Upsilon_j]$, where

$$\begin{aligned}\mathbf{A}_{clj} &= \mathbf{A}_{sj} + \mathbf{B}_{sj}\mathbf{K}_s, \quad \mathbf{B}_{ILCj} = \begin{bmatrix} \mathbf{B}_j N \\ 1 \end{bmatrix} \\ \Gamma_j &= \mathbf{C}_{cl}\mathbf{A}_{clj}^r, \quad \Upsilon_j = \mathbf{C}_{cl}\mathbf{A}_{clj}^{r-1}\mathbf{B}_{ILCj}\end{aligned}\quad (23)$$

is, in general, nonconvex due to the presence of powers of the matrix $\mathbf{A}_{cl}(\lambda)$.

There are many tools available to form a convex set for given vertices as in (23). Here, MATLAB compatible geometric bounding software is used and gives

$$\begin{aligned}\tilde{X} &= [\hat{\mathbf{A}}_{cl}(\lambda) \ \hat{\mathbf{B}}_{ILC}(\lambda) \ \hat{\Gamma}(\lambda) \ \hat{\Upsilon}(\lambda)] \\ \tilde{X} &= \sum_{j=1}^{\mathcal{M}} \lambda_j [\hat{\mathbf{A}}_{clj} \ \hat{\mathbf{B}}_{ILCj} \ \hat{\Gamma}_j \ \hat{\Upsilon}_j].\end{aligned}\quad (24)$$

As a result, (21) becomes

$$\begin{aligned}\eta_{k+1}(p+1) &= \hat{\mathbf{A}}(\lambda)\eta_{k+1}(p) + \hat{\mathbf{B}}_0(\lambda)e_k(p) \\ e_{k+1}(p) &= \hat{\mathbf{C}}(\lambda)\eta_{k+1}(p) + \hat{D}_0(\lambda)e_k(p)\end{aligned}\quad (25)$$

with the uncertain matrices

$$\begin{aligned}\hat{\mathbf{A}}(\lambda) &= \hat{\mathbf{A}}_{cl}(\lambda), \quad \hat{\mathbf{B}}_0(\lambda) = \hat{\mathbf{B}}_{ILC}(\lambda)K_3 \\ \hat{\mathbf{C}}(\lambda) &= -\hat{\Gamma}(\lambda), \quad \hat{D}_0(\lambda) = 1 - \hat{\Upsilon}(\lambda)K_3\end{aligned}\quad (26)$$

and the set of vertices which have the required convexity property. The state-space model (25) is that of a discrete linear repetitive process [7] and Sections IV-A and IV-B uses the stability theory of these processes to undertake a robust ILC design.

A. Robust Stability Analysis Along the Trial

A stability theory for linear constant pass length processes has been developed in a Banach space setting [7]. This stability theory requires that a bounded initial pass profile produces a bounded sequence of pass profiles (in k) either over the finite and fixed pass length or, in stronger form, independent of the pass length. The extension to control law design has been the subject of much research leading to computationally feasible design algorithms. In this brief, the stability along the pass property is used and to conform with the vast majority of the ILC literature, the word pass is replaced by trial from this point onward. A standing result is that if stability along the trial holds for a linear repetitive process, the ILC dynamics represented in this form have monotonic trial-to-trial error convergence [7].

Introduce the notation

$$\Phi(\lambda) = \begin{bmatrix} \hat{\mathbf{A}}(\lambda) & \hat{\mathbf{B}}_0(\lambda) \\ \hat{\mathbf{C}}(\lambda) & \hat{D}_0(\lambda) \end{bmatrix}, \quad \xi(k, p) = \begin{bmatrix} \eta_{k+1}(p) \\ e_k(p) \end{bmatrix}\quad (27)$$

and $\mathcal{P}(\lambda) = \text{diag}(\mathbf{P}_1(\lambda), \mathbf{P}_2(\lambda))$. Then by linear repetitive process stability theory, (25) is stable along the trial [7] if

$$\Phi^T(\lambda)\mathcal{P}(\lambda)\Phi(\lambda) - \mathcal{P}(\lambda) < 0. \quad (28)$$

B. Robust ILC Design Using the Guaranteed Cost Approach

The design problem considered in this section is to determine gain K_3 that guarantees "fast" trial-to-trial error convergence. If the LMIs resulting from (28) are feasible, there exist infinitely many solutions but no criteria for selecting them to reflect the expected performance. One way of avoiding this obstacle is to use the guaranteed cost method introduced for standard linear systems in [6] and for repetitive processes in [7]. Since the dynamics along the trial of the ILC system (25) depend on (12), the following cost function is used:

$$\begin{aligned}J_{ILC} &= \sum_{k=0}^{k^*} \sum_{p=0}^{\alpha-1} e_{k+1}^2(p) \\ &= \sum_{k=0}^{k^*} \sum_{p=0}^{\alpha-1} \xi^T(k, p) \mathbf{U}^T(\lambda) \mathbf{U}(\lambda) \xi(k, p)\end{aligned}\quad (29)$$

with $\mathbf{U}(\lambda) = [\hat{\mathbf{C}}(\lambda) \ \hat{D}_0(\lambda)]$, where k^* denotes the number of trials to be completed. Hence, the requirement for stability along the trial becomes

$$\Phi^T(\lambda)\mathcal{P}(\lambda)\Phi(\lambda) - \mathcal{P}(\lambda) + \mathbf{U}^T(\lambda)\mathbf{U}(\lambda) \leq 0. \quad (30)$$

The following is the main result of this brief.

Theorem 1: Consider uncertain discrete ILC dynamics described by (25). Then, the property of robust stability along the trial holds and the associated cost function J_{ILC} (29) satisfies the upper bound in [5, Th. 2] if, for given boundary conditions, $\eta_{k+1}(0) = \mathbf{f}$, $e_0(p) = g$ there exist compatibly dimensioned matrices $\mathbf{G}_1$ and $\mathbf{Y}_{1j} > 0$ and scalars G_2 , W , $Y_{2j} > 0$, $\beta_1 > 0$, $\beta_2 > 0$ such that $\forall j = 1, \dots, \mathcal{M}$

$$\begin{bmatrix} \mathbf{G}_1 + \mathbf{G}_1^T - \mathbf{Y}_{1j} & (*) & (*) & (*) & (*) \\ \mathbf{0} & 2G_2 - Y_{2j} & (*) & (*) & (*) \\ \hat{\mathbf{A}}_{clj}\mathbf{G}_1 & \hat{\mathbf{B}}_{ILCj}W & \mathbf{Y}_{1j} & (*) & (*) \\ -\hat{\Gamma}_j\mathbf{G}_1 & G_2 - \hat{\Upsilon}_jW & \mathbf{0} & Y_{2j} & (*) \\ -\hat{\Gamma}_j\mathbf{G}_1 & G_2 - \hat{\Upsilon}_jW & \mathbf{0} & 0 & 1 \end{bmatrix} \geq 0 \quad (31)$$

$$\begin{bmatrix} \beta_1 & \mathbf{f}^T \\ \mathbf{f} & \mathbf{Y}_{1j} \end{bmatrix} \geq 0, \quad \begin{bmatrix} \beta_2 & g \\ g & Y_{2j} \end{bmatrix} \geq 0. \quad (32)$$

Then, gain K_3 of the correction term (17) is given by

$$K_3 = W/G_2. \quad (33)$$

The proof follows the arguments of [5], [7], and [8], where a parameter-independent Lyapunov function was used.

The following minimization procedure can be applied to this last result:

$$\begin{aligned}\min & (\beta_1 + \beta_2) \\ \text{s. t.:} & \text{diag}(\mathbf{Y}_{1j}, Y_{2j}) > 0, (31) \text{ and } (32)\end{aligned}\quad (34)$$

for $j = 1, \dots, \mathcal{M}$. If a solution exists, K_3 in (17) is given by (33).

V. EXPERIMENTAL VALIDATION

A. Equipment Details and Control Design

The robust ILC design developed in Section IV has been applied experimentally to position tracking of a PMSM (drive A) mechanically connected to a second PMSM (drive B), which is used to generate an external disturbance in the form of a load torque. Between these two motors, an additional rotary mass can be inserted to evaluate the robustness of the design to changes in the mass moment of inertia. The angular positions of both motor shafts are measured by resolvers. The resolver digital converters, used in both drives, have a resolution of $2\pi/2^{14}$ rad and an accuracy of ± 0.007 rad. The angular velocities are calculated numerically as a 16-point moving average of the position differences with the same sampling time as the current control loop. The underlying current control loops of both PMSMs are based on a field-oriented control approach. This corresponds to decoupling of the orthogonal current components and nonlinear feedback linearization [9]. A detailed description of the experimental setup is given in [5].

The dynamic model for the control of the angular position is

$$\begin{aligned} T_e^A(t) &= i_q^{\text{ref}A}(t) k_t^A \\ &= J_M \frac{d^2 \theta_M^A(t)}{dt^2} + B_M \frac{d\theta_M^A(t)}{dt} + T_l^B(t) \end{aligned} \quad (35)$$

where T_e^A denotes the electromagnetic torque generated by drive A, $i_q^{\text{ref}A}$ denotes the reference motor current of drive A, k_t^A denotes the corresponding torque constant, J_M denotes the overall mass moment of inertia, B_M denotes the resulting friction coefficient, θ_M^A denotes the motor shaft angle, and T_l^B denotes the load torque generated by drive B. The resulting continuous-time state-space model for $T_l^B = 0$ is

$$\begin{aligned} \dot{x}(t) &= \mathbf{A}_c x(t) + \mathbf{B}_c u(t) \\ y(t) &= \mathbf{C}_c x(t) \end{aligned} \quad (36)$$

where

$$x(t) = \begin{bmatrix} \theta_M^A(t) \\ \omega_M^A(t) \end{bmatrix}, \quad u(t) = i_q^{\text{ref}A}(t) \quad (37)$$

and

$$\begin{aligned} \mathbf{A}_c &= \begin{bmatrix} 0 & 1 \\ 0 & -\frac{B_M}{J_M} \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ \frac{k_t^A}{J_M} \end{bmatrix} \\ \mathbf{C}_c &= \begin{bmatrix} 1 & 0 \end{bmatrix}. \end{aligned} \quad (38)$$

In this representation, the angular velocity of the motor shaft is $\omega_M^A(t) = (d\theta_M^A(t)/dt)$.

In the remainder of this brief, the model uncertainty is given as follows.

- 1) The mass moment of inertia varies between a minimum value $J_{M,\min} = 5.9 \cdot 10^{-4} \text{ kg} \cdot \text{m}^2$ and a maximum $J_{M,\max} = 10.6 \cdot 10^{-4} \text{ kg} \cdot \text{m}^2$.
- 2) The friction coefficient varies between a minimum value $B_{M,\min} = 1.3 \cdot 10^{-3} \text{ kg} \cdot \text{m}^2/\text{s}$ and a maximum $B_{M,\max} = 1.5 \cdot 10^{-3} \text{ kg} \cdot \text{m}^2/\text{s}$.

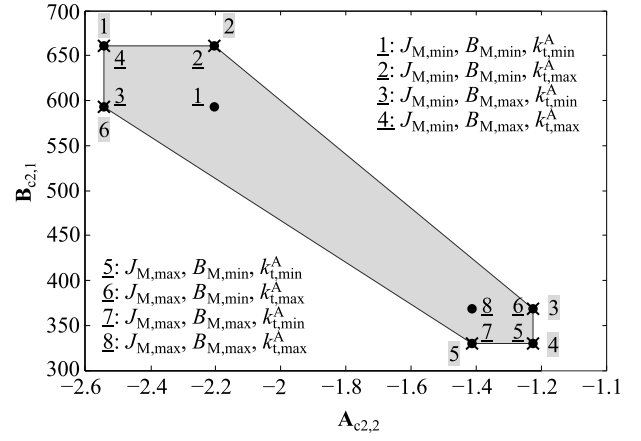


Fig. 2. Vertices of the continuous-time plant model (36) marked on the plane formed by the element (2, 2) of the state matrix $\mathbf{A}_c$ and the element (2, 1) of the input vector $\mathbf{B}_c$.

- 3) The torque constant varies between a minimum value $k_{t,\min}^A = 0.35 \text{ N} \cdot \text{m/A}$ and a maximum $k_{t,\max}^A = 0.39 \text{ N} \cdot \text{m/A}$.

Given these bounds on the parameters in (36), a set of eight vertices marked by dots in Fig. 2 can be constructed. A minimal convex set containing these vertices consists of the six vertices marked by x . Alternatively, a MATLAB compatible geometric bounding software may be used to find the six vertices of a minimum convex set.

The new design in this brief requires a discrete-time system representation, which is obtained using the Euler discretization method preserving the convexity of the uncertain continuous-time state-space model. For the six vertices defining the minimum convex set of the continuous-time model (Fig. 2), the corresponding discrete-time state-space models $[\mathbf{A}_j, \mathbf{B}_j]$ were calculated for the sampling time $T_{\text{sm}}^A = 2.5 \cdot 10^{-3} \text{ s}$. The output vector of each of them is $\mathbf{C} = [1 \ 0]$. The relative degree for all of these discrete models is $r = 2$.

The MATLAB toolboxes for specifying convex programs (CVX) and semidefinite-quadratic-linear programming (SDPT3) have been used to solve the minimization problem (12), where the quadratic cost function is defined by the following choices for the weighting matrices $\mathbf{Q} = \text{diag}(1.8 \cdot 10^4, 0.1, 4)$, $R = 0.9$, and $\boldsymbol{\gamma} = [0.01 \ 0 \ 0]^T$. Since the strict inequality $\mathbf{Y} > 0$ cannot be implemented in CVX, it has been replaced by $\mathbf{Y} \geq 10^{-10} \cdot \mathbf{I}$. The numerical solution of the control problem leads to

$$\mathbf{K}_s = \begin{bmatrix} -99.4484 & -0.9270 & 1.3916 \end{bmatrix} \quad (39)$$

and hence, using (7), $\mathbf{K}_1$ and $\mathbf{K}_2$.

The design of both the static and dynamic feedforward controllers is based on the parameters of the closed-loop system. As detailed in Section III, this is done for $\mathbf{K}_1$ arising from (39) and $\mathbf{K}_2 = 0$. Moreover, the average values of the system parameters are used, i.e., $J_M = 8.25 \cdot 10^{-4} \text{ kg} \cdot \text{m}^2$, $B_M = 1.4 \cdot 10^{-3} \text{ kg} \cdot \text{m}^2/\text{s}$, and $k_t^A = 0.37 \text{ N} \cdot \text{m/A}$. The application of Euler's discretization method ($T_{\text{sm}}^A = 2.5 \cdot 10^{-3} \text{ s}$) gives the discretized state-space matrices

$$\mathbf{A}_{\text{av}} = \begin{bmatrix} 1 & 0.0025 \\ 0 & 0.9958 \end{bmatrix}, \quad \mathbf{B}_{\text{av}} = \begin{bmatrix} 0 \\ 1.1212 \end{bmatrix}$$

and $\mathbf{C}_{av} = [1 \ 0]$. The state-space model of the closed-loop feedback system used to design both feedforward controllers has the form (see Fig. 1)

$$\begin{aligned} x_{k+1}(p+1) &= (\mathbf{A}_{av} + \mathbf{B}_{av}\mathbf{K}_1)x_{k+1}(p) + \mathbf{B}_{av}w_{k+1}(p) \\ y_{k+1}(p) &= \mathbf{C}_{av}x_{k+1}(p). \end{aligned} \quad (40)$$

The dc gain of this system is 0.01005547 and hence $N = 99.4484$.

The closed-loop system (40) has no zeros and therefore the dynamic feedforward controller is designed using the perfect tracking algorithm [10], which provides signal $f(p)$ (see Fig. 1) as

$$f(p) = \frac{1 - 0.9564 z^{-1} + 0.2352 z^{-2}}{0.0028} y^{\text{ref}}(p+r). \quad (41)$$

Since the relative degree of the plant models is $r = 2$, the ILC law (16) in this case is

$$v_{k+1}(p) = v_k(p) + K_3 e_k(p+2). \quad (42)$$

The application of (42) may lead to a nonconvex ILC model (21) due to the form of $\mathbf{\Gamma}(\lambda)$ and $\mathbf{\Upsilon}(\lambda)$ given by (22); therefore, MATLAB compatible geometric bounding software has been used to find the six vertices of a minimum convex set (24) for this example. Then, CVX and SDPT3 have been employed to solve the minimization problem (34) for $\mathbf{f} = [0.01 \ 0 \ 0]^T$ and $g = 0.1$. The strict inequalities $\text{diag}(\mathbf{Y}_{1j}, \mathbf{Y}_{2j}) \succ 0$ of (34) have been replaced by $\text{diag}(\mathbf{Y}_{1j}, \mathbf{Y}_{2j}) \geq 10^{-10} \cdot \mathbf{I}$ for all $j = 1, \dots, 6$. Finally, the ILC design yields the gain $K_3 = 0.6942$.

B. Experimental Results and Discussion

To counteract high-frequency noise and quantization errors in the implementation, an off-line low-pass filtering of the tracking error has been implemented at the end of each trial. For this purpose, a zero-phase digital filter, available in MATLAB through the function `filtfilt`, was used. This filter is parameterized as a second-order low-pass digital Butterworth filter with a cutoff frequency of 30 Hz and sampling period T_{sm}^A .

In the remainder of this section, the experimental results obtained for three control strategies are given to demonstrate the capabilities of the new ILC design. The following alternatives are investigated and compared with each other:

- 1) the new ILC structure shown in Fig. 1;
- 2) the simplified, parallel ILC structure where an ILC signal acts only on the plant input (switch S is in the position 0);
- 3) an ILC structure without integral action.

These experiments include the impact of *a priori* unknown disturbances and also model uncertainty. The disturbances are applied as variations in the load torque, whereas model uncertainty is addressed by changing the mass moment of inertia after a particular number of trials. Moreover, the effects of changes in the reference trajectory are also investigated. Table I and Fig. 3 give the parameters and signals used. In all experiments, the torque constant and the friction coefficient are held constant. Hence, the intervals for these parameters reflect

TABLE I
VARIATION OF THE REFERENCE SIGNAL, THE LOAD TORQUE SIGNAL, AND THE MASS MOMENT OF INERTIA DURING THE ROBUST ILC APPLICATION

Trials	$y^{\text{ref}}(p)$	$T_l^B(p)$	J_M [kg·m ²]
1 – 30	$y_1^{\text{ref}}(p)$	0	$5.9 \cdot 10^{-4}$
31 – 60	$y_1^{\text{ref}}(p)$	$3k_t^A \mathbb{I}(p-100)$	$5.9 \cdot 10^{-4}$
61 – 90	$y_1^{\text{ref}}(p)$	$3k_t^A \mathbb{I}(p-100)$	$10.6 \cdot 10^{-4}$
91 – 500	$y_2^{\text{ref}}(p)$	$2k_t^A \mathbb{I}(p-250)$	$8.6 \cdot 10^{-4}$

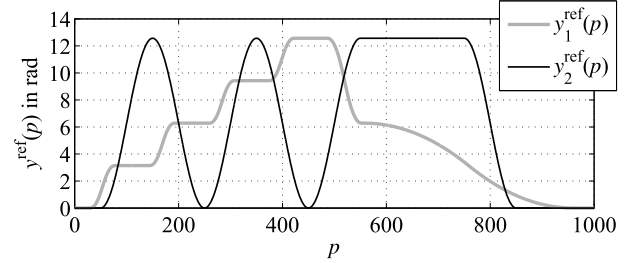


Fig. 3. Reference trajectories during the robust ILC.

the remaining uncertainty in the experimental identification of the corresponding numerical parameter values.

In the computation of the ILC laws, at each sampling instant of the current trial, the tracking error contribution from the previous trial is shifted by $r = 2$ samples [see (42)]. Hence, the last two samples of the ILC signal, i.e., $v_{k+1}(999)$ and $v_{k+1}(1000)$, cannot be implemented. Instead, the following assignments were made: $e_{k+1}(1001) = e_{k+1}(1000)$ and $e_{k+1}(1002) = e_{k+1}(1000)$. Hence, the first two samples of the tracking error from the previous trial are not used and in each case replaced by zero.

1) *Evaluation of the Proposed ILC Scheme ($S = 1$):* Assuming zero boundary conditions, the new ILC structure shown in Fig. 1 has been experimentally tested according to the scenario in Table I. The root-mean-square (RMS) tracking error for the experimental data is shown in Fig. 4, where the corresponding plot is denoted by “ILC Fig. 1”. These results show that the new ILC implementation needs four trials to result in very small tracking errors that are close to the RDC resolution—despite all the investigated changes in the load torque, the mass moment of inertia, and the reference trajectory. Moreover, the integral action of the state feedback controller reduced the steady-state tracking error caused by the disturbance torque starting immediately during the trial in which the disturbance appeared.

2) *Evaluation of a Simplified, Parallel ILC Structure ($S = 0$):* A simplification of Fig. 1 occurs when the ILC signal $v_{k+1}(p)$ is connected only to the system input, which makes the structure parallel. An identical set of experiments was performed for this alternative ILC scheme. The resulting RMS tracking error for the first 150 executed trials is depicted in Fig. 4 denoted by “ILC simplified”. For this ILC structure, a slower trial-to-trial error convergence occurs after adding or changing load disturbances, where 14 trials, as opposed to only 4, are required to obtain tracking errors

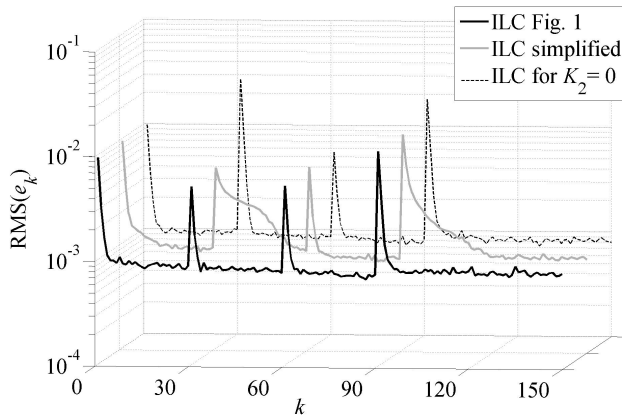


Fig. 4. RMS values of the tracking error for the first 150 trials.

close to the RDC resolution. This effect is related to changes in the integral action output signal $\psi_{k+1}(p)$ until perfect tracking is obtained again.

3) *Evaluation of the ILC Structure Without Integral Action* ($K_2 = 0$): The ILC schemes considered above are equivalent for $K_2 = 0$, i.e., when the integral action is disabled. A set of experiments was also performed for this case. The resulting RMS tracking error is denoted by “ILC for $K_2 = 0$ ” and shown in Fig. 4. This set of experiments is of particular interest in assessing the tracking error after adding a load torque, on trial $k = 31$. In this case, a steady-state tracking error occurs that can only be reduced/counteracted by the ILC after the next trial has begun. The RMS tracking error is significantly larger at trial $k = 31$ than in both previous sets of experiments, justifying the inclusion of the integral action.

VI. CONCLUSION

In this brief, an innovative ILC law, augmented by a state feedback controller with integral action, has been developed and applied to the angular position tracking of a PMSM. The integral action included in the state feedback controller reduces the steady-state error caused by a load disturbance torque during the trial in which it appears. The new robust ILC design does not require the storage of the additional signal which is the product of the gain vector and the state vector along the trials. It can be used for plants with relative degree $r \geq 1$ and also in the case of uncertainty in the plant model parameters.

The results confirm that the new ILC design is capable of achieving small tracking errors after a small number of trials (four trials for the considered PMSM), including the case when model uncertainty, unmeasured disturbances, and measurement noise are present.

Given the results and the positive experimental validation reported in this brief, there are a number of areas for future research. These include implementation on more sophisticated drive train topologies, e.g., those involving elasticity. Also the ILC design in the presence of noise should be investigated. The analysis and experimental results in this brief are for SISO systems. Extension of the theory to the case of square MIMO systems should be considered. In the PMSM application, the state vector entries can be directly measured. If this is not the case in other applications, an additional observer is required.

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